Controlling Product Costing (CO-PC I)

This case study explains an integrated product costing process in details and thus fosters a thorough understanding of each process step and underlying SAP functionality.

Product
S/4HANA 2023
Global Bike

Fiori 3.0

Level
Undergraduate
Graduate
Beginner

Focus
Product Costing

Author
Michael Boldau
Stefan Weidner

Version
4.3

Last Update
July 2025

MOTIVATION

The data entry requirements in the controlling exercises (CO 1 through CO 3) were minimized because most of the data was stored in the SAP system. This stored data, known as master data, simplifies the processing of business transactions.

In this case study, we create a variant of an existing product in order to calculate associated costs. The cost estimate for this new bike version was initiated by the sales department, which plans on running a sales campaign with a special edition of this bike. After the new material is created in the system, existing bills of materials and routings need to be copied and modified.

After the material has been created the product cost estimate is calculated.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to work successfully through this case study, it is not necessary to have finished the CO exercises. Anyway, it is recommended.

NOTES

This case study uses the Global Bike (GB) data set.

SAP University Alliances

© SAP UCC Magdeburg

Process Overview

Learning Objective Understand and perform an integrated product costing process.

Time 65 min

Scenario In order to perform a product costing process you will take on different roles within the Global Bike company, e.g. production data administrator, cost accountant or chief accountant. Overall, you will work in the Production Planning (PP) and the Controlling (CO) departments.

Employees involved Jermain Kumins (Shop Floor Worker 1)

Jamie Shamblin (Controller)

Shuyuan Chen (Head of Accounting)

You start the product costing process by creating a slightly changed version of your top-selling Deluxe Touring bike. As you plan to manufacture and sell a white color in the future, you create new finished goods and necessary raw materials. The bill of material and routing are changed accordingly. Prices are calculated for the new frame and then released.

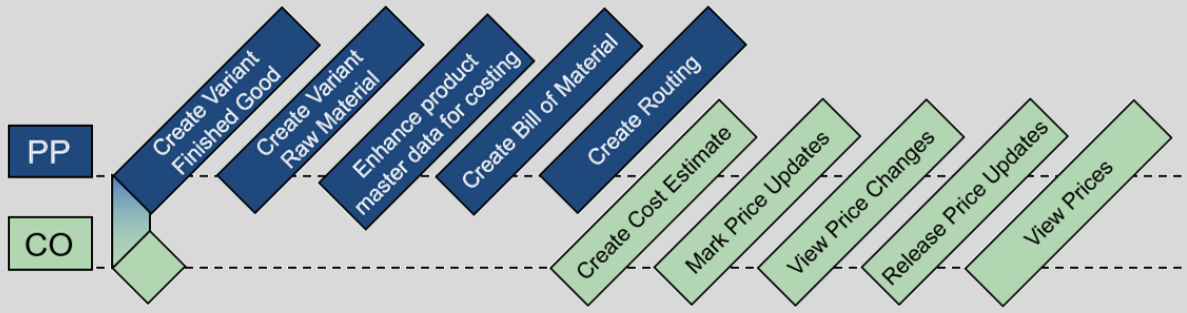


Table of Contents

Process Overview 2
Step 1: Create Variant Finished Good 4
Step 2: Create Variant Raw Material 7
Step 3: Extend Raw Material for Calculation 10
Step 4: Create Bill of Material 12
Step 5: Create Routing 14
Step 6: Create Cost Estimate 17
Step 7: Mark Price Updates 20
Step 8: View Prices Changes 22
Step 9: Release Price Updates 24
Step 10: View Prices 25
CO-PC I Challenge 27

Step 1: Create Variant Finished Good

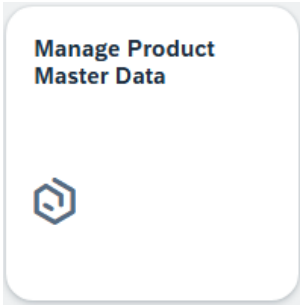
Task Copy the product master data of a finished product. **Time** 10 min

Short Description Create a variant of the Deluxe Touring Bike (black) for product costing. In order to do so, you should create a new product while copying the data from your Deluxe Touring Bike (black).

Name (Position) Jermain Kumins (Shop Floor Worker 1)

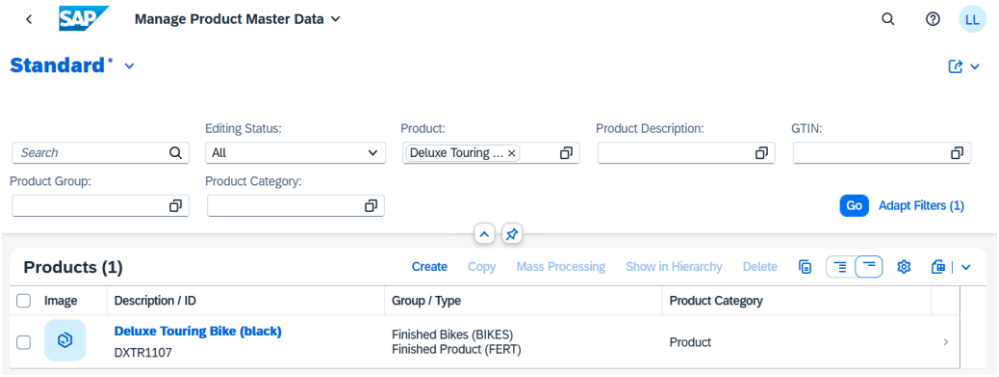
To create the product master record for a finished product, use the *Manage Product Master Data* app in the *Controlling* area on the *Product Cost Controlling* page in the *Shop Floor Worker* role.

Start

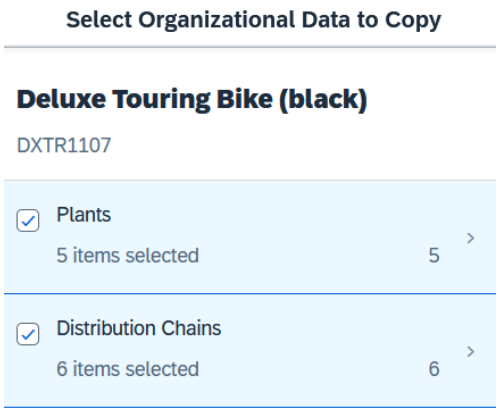


Enter as *Product* **DXTR1###** (replace ### with your three-digit number) and choose Go.

DXTR1###

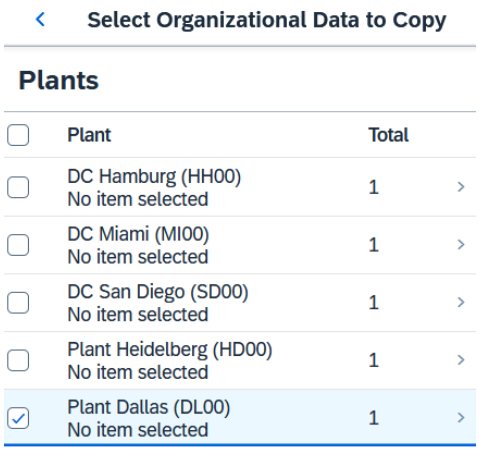


Select the line by clicking the checkbox at the beginning of the line and click Copy. In the following popup, in the Plants row on the far right, select the icon ›.



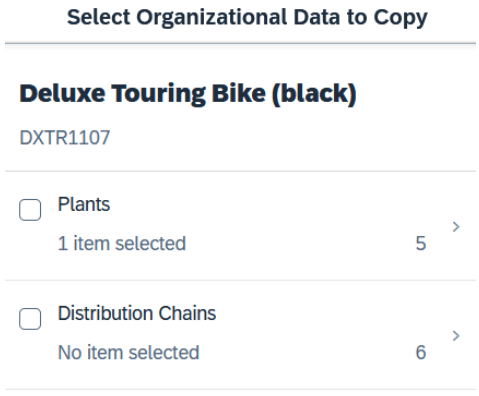
In the following window, limit the selection to the *plant* **DL00** plant.

DL00



To return to the previous menu, press  . Then deselect *Distribution Chains* line.

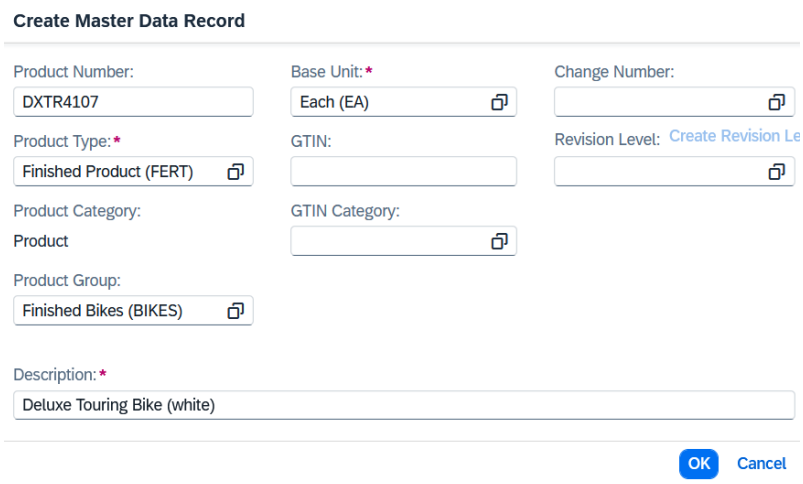
~~Distribution Chains~~



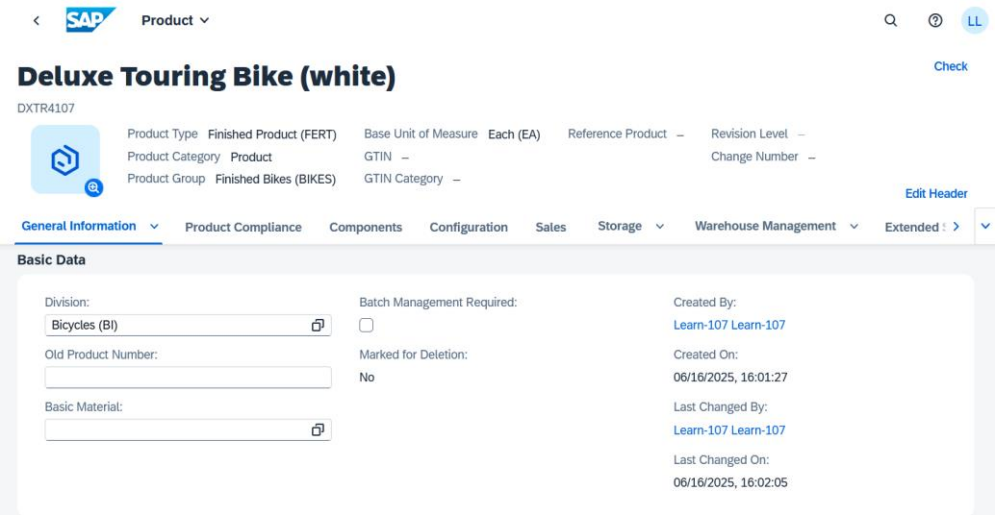
Confirm your Entries with  .

In the Pop-Up *Create Master Data Record* enter as *Product Number* **DXTR4###**. *Product Type, Product Group* and *Base Unit* are already filled. Change the *Description* to **Deluxe Touring Bike (white)**.

DXTR4###.
Deluxe Touring Bike (white)

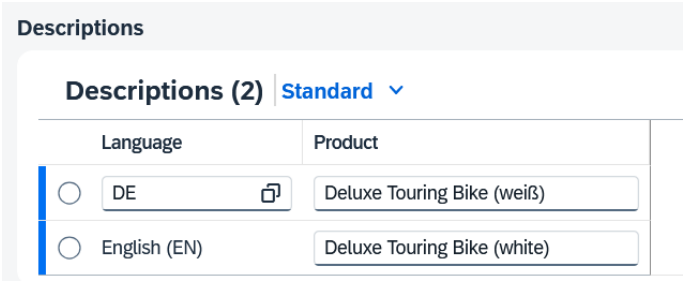


Confirm your Entries with OK. You will now see the detailed information of the new product.



In the *Descriptions* section, you change the product description in the language *DE (German)* to **Deluxe Touring Bike (weiß)**. If you speak Russian or French, you can change the descriptions as well or delete the entries otherwise.

Deluxe Touring Bike (weiß)



Click on Create to save your product.

Click on SAP to return to the SAP Fiori Launchpad.

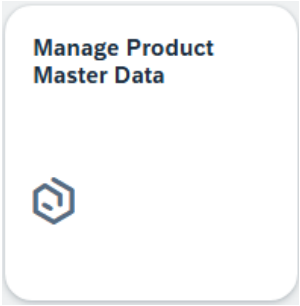
Step 2: Create Variant Raw Material

Task Copy a new raw material for your bike. **Time** 5 min

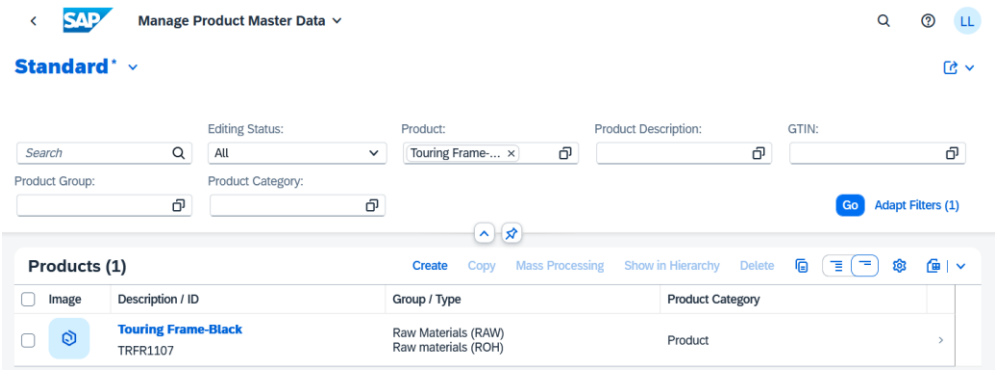
Short Description Create a white frame as a new material. It will be used later in the BOM of your new Deluxe Touring Bike (white)

Name (Position) Jermain Kumins (Shop Floor Worker1)

To create the product master record for a finished product, use the *Manage Product Master Data* app in the *Controlling* area on the *Product Cost Controlling* page in the *Shop Floor Worker* role. Start

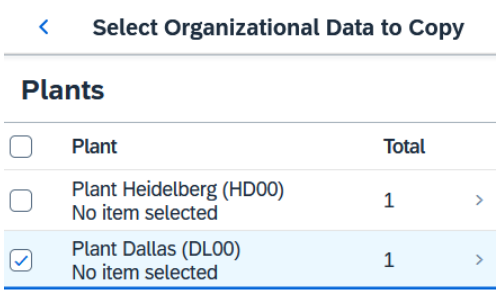


Enter as *Product* **TRFR1###** (replace ### with your three-digit number) and choose Go. TRFR1###



Select the line by clicking the checkbox at the beginning of the line and click Copy. In the following popup, in the Plants row on the far right, click on the button.

In the following window, limit the selection to the *plant* **DL00** plant. DL00

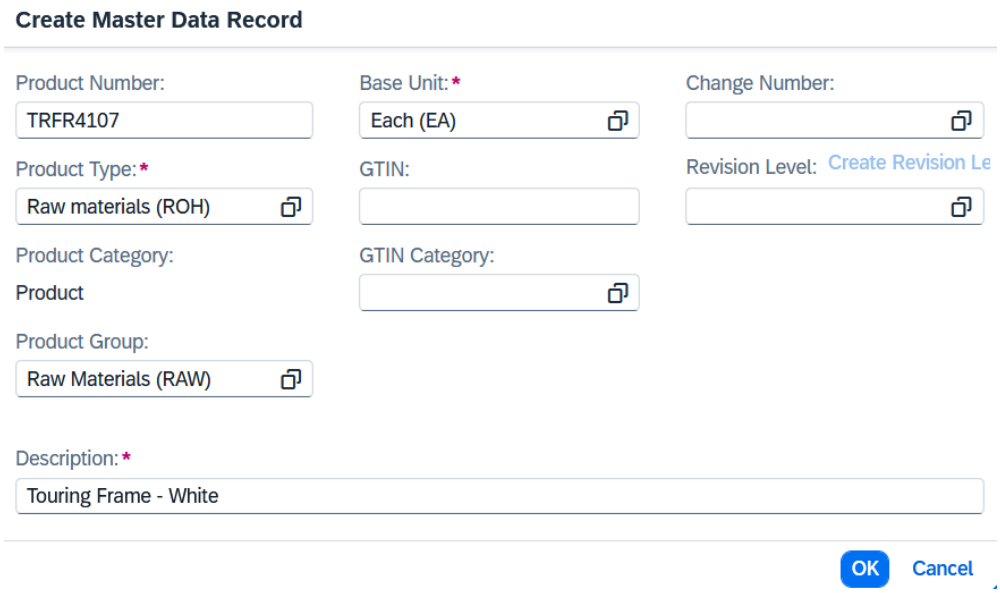


To return to the previous menu, press <. Confirm your Entries with OK.

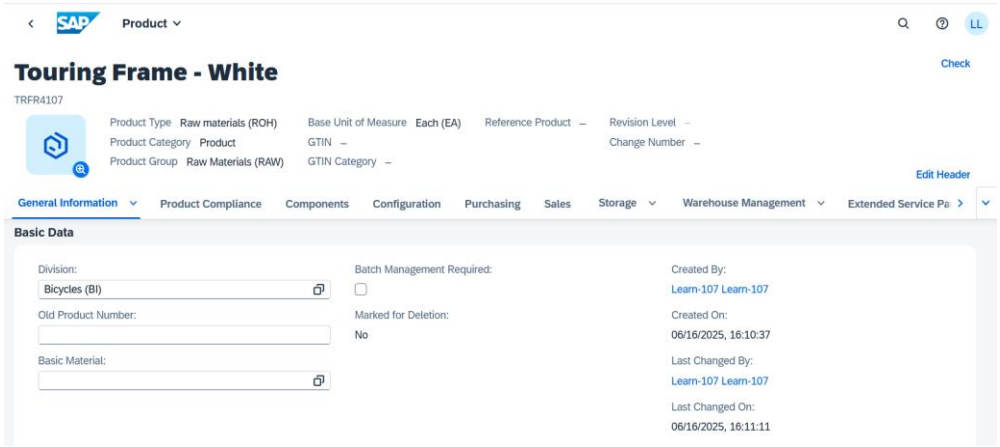
In the Pop-Up *Create Master Data Record* enter as *Product Number* **TRFR4###**. *Product Type, Product Group* and *Base Unit* are already filled. Change the *Description* to **Touring Frame - White**.

TRFR4###

Touring Frame - White

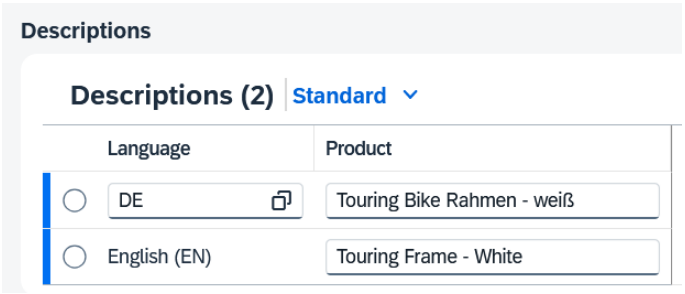


Confirm your Entries with OK. You will now see the detailed information of the new product.



In the *Descriptions* section, you change the product description in the language *DE (German)* to **Touring Bike Rahmen - weiß**. If you speak Russian or French, you can change the descriptions as well or delete the entries otherwise.

Touring Bike Rahmen - weiß



Click on Create to save your product.

Click on SAP to return to the SAP Fiori Launchpad.

Step 3: Extend Raw Material for Calculation

Task Enhance your raw material master data. **Time** 5 min

Short Description In order to enable the cost calculation with quantity structure, the previously created master record must be extended

Name (Position) Jermain Kumins (Shop Floor Worker 1)

To extend the material master record for your raw material, use the *Change Material* app in the *Controlling* area on the *Product Cost Controlling* page in the *Shop Floor Worker* role.

Start

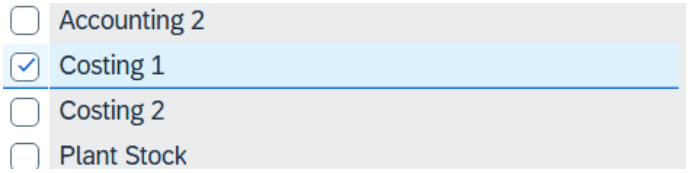


Enter as *Material* **DXTR4###** and press Continue. You will enter the view selection.

DXTR4###

Choose view *Costing 1* and confirm with ✓.

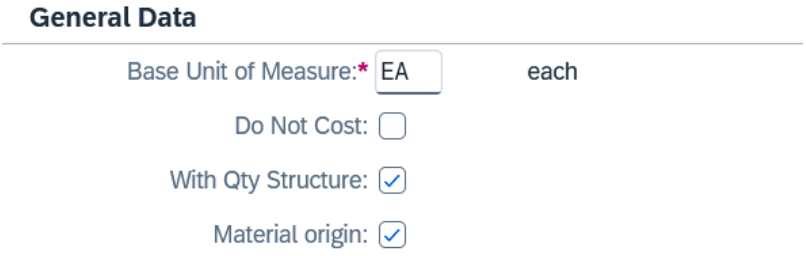
Costing 1



Enter *Plant* **DL00** and confirm with ✓.

DL00

In the *General Data* section, select the Checkboxes *With Qty Structure:* and *Material origin:*



Choose Save afterwards. You will get the following confirmation:



Click on  to return to the SAP Fiori Launchpad.

Step 4: Create Bill of Material

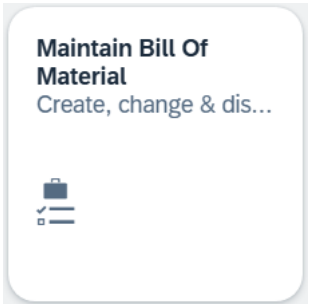
Task Create a new BOM. **Time** 10 min

Short Description Create a new BOM for your Deluxe Touring Bike (white) to be used in the product costing process.

Name (Position) Jermain Kumins (Shop Floor Worker 1)

To create a bill of material, use the *Maintain Bill of Material* app in the *Controlling* area on the *Product Cost Controlling* page in the *Shop Floor Worker* role.

Start

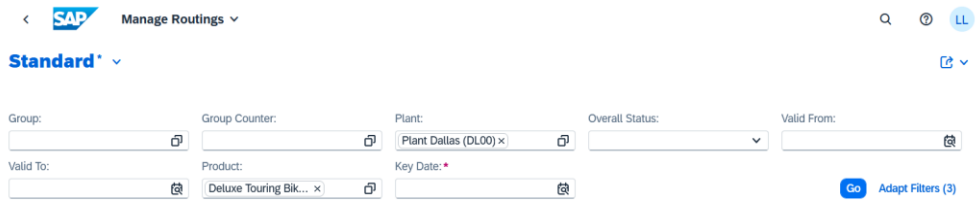


A *bill of material* contains all the components that go into the manufacture of a product. Bills of material are used in material requirements planning, production, procurement and product costing. Furthermore, alternative BOMs can be used to represent the same product by several BOMs. This is used, for example, to represent different manufacturing processes.

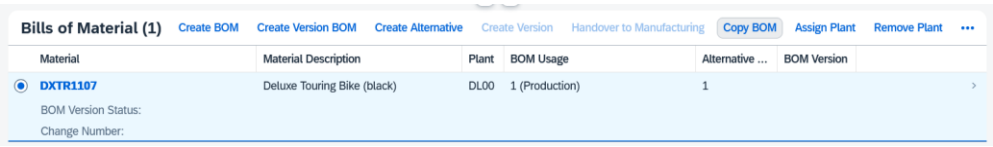
Bill of Material

On the screen *Maintain Bill of Material* enter in the field *Material:* **DXTR1###** and in the field *Plant*, enter **DL00**.

DXTR1###
DL00

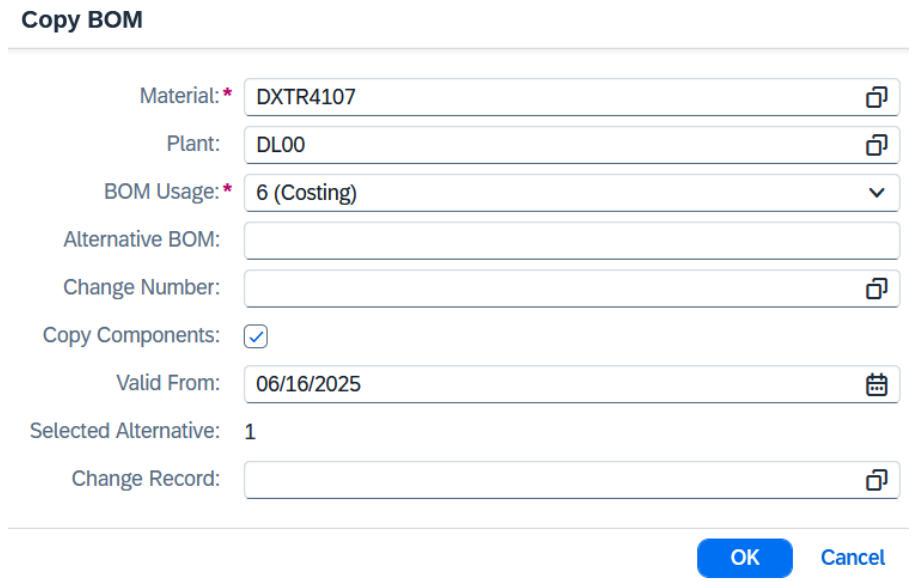


Press Go to find the corresponding bill of material to your product. Select the entry and choose Copy BOM.

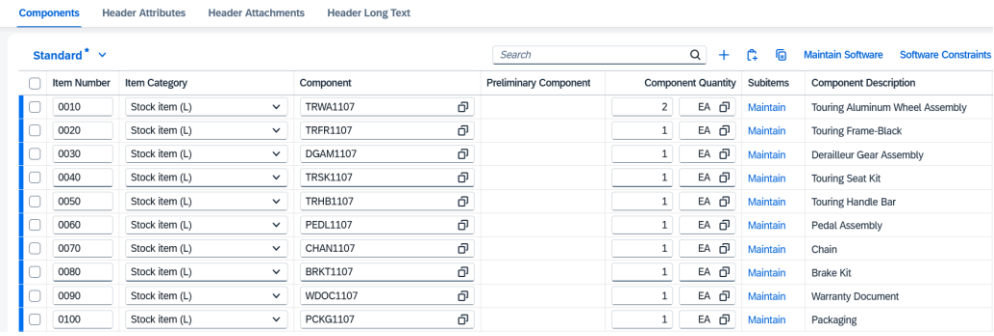


In the *Copy BOM* pop-up, enter **DXTR4###** in the *Material* field and **DL00** in the *Plant* field, and select **6 (Costing)** in the *BOM Usage* field.

DXTR4###
DL00
6 (Costing)



Click on OK to apply your entries. You will get to the new screen *Maintain Bill of Material*.

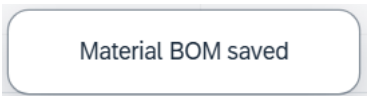


Change the *Component* in Item Number *0020* to your previously created frame **TRFR4###** and press Enter.

TRFR4###



Click on Create. You will get a confirmation from the system.



Click on SAP to return to the SAP Fiori Launchpad.

Step 5: Create Routing

Task Create a new routing. **Time** 5 min

Short Description Create a new routing for your Deluxe Touring Bike (white) to be used later in the product costing process.

Name (Position) Jermain Kumins (Shop Floor Worker 1)

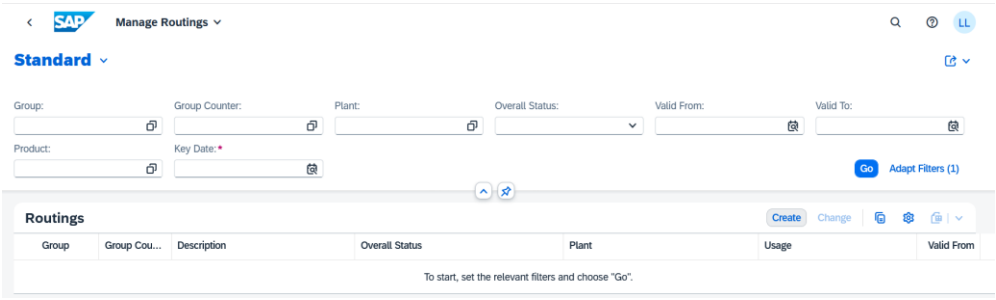
To create the routing, use the app *Create Routing* in the *Controlling* area on the *Product Cost Controlling* page in the *Shop Floor Worker* role.

Start



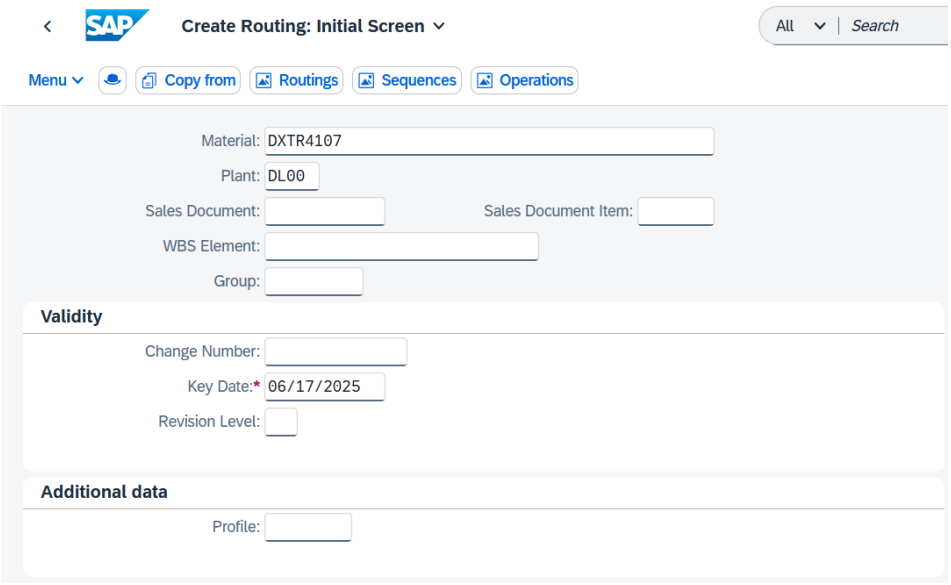
Note Routings contain the steps that are necessary for production and these steps are called operations. Routings include the sequence in which the operations occur, and the work centers at which the operations are to be executed.

In the screen *Manage Routing*, click on Create.



Enter your *Product* **DXTR4###** and plant **DL00** and make sure the *Group* field is left empty.Click on Copy from.

DXTR4###
DL00



Make sure that **Routing** is selected on the *Select Type* pop up.

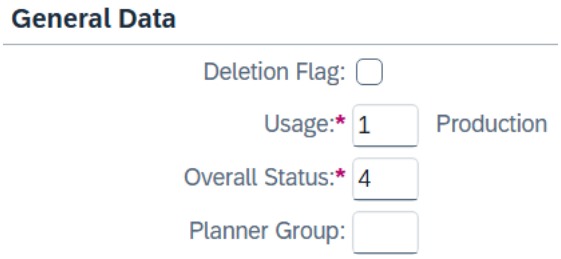
Routing

Klick on (*Continue*). On the next screen *Create Using Copy Function: Select Object to be Copied* enter the *Material* **DXTR1###**. As *Plant* **DL00** should already be entered. If not, enter it.

DXTR1###
DL00

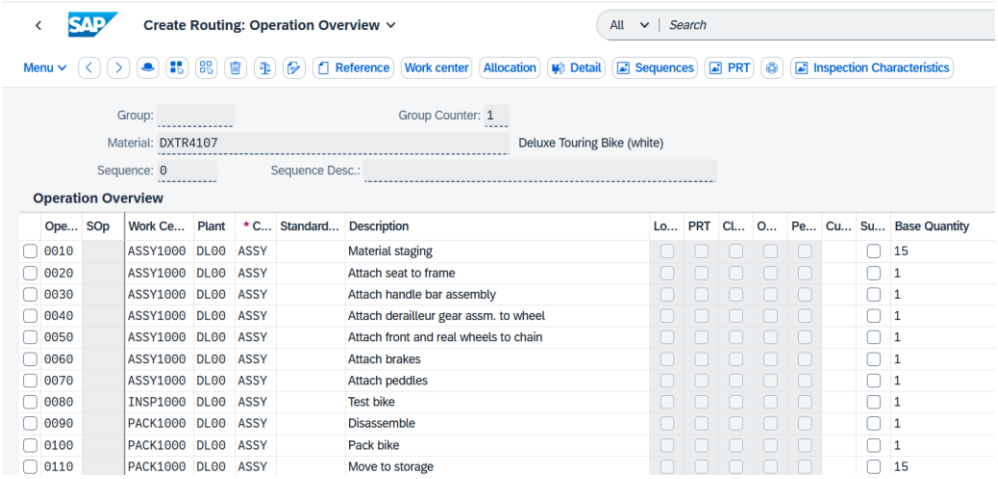
Again, click on (*Continue*). In the next screen *Create Routing: Header Data Check* set the *Overall Status:* to **4** (Released (General)).

4



Press *Enter* to confirm your changes. The system now shows you in the *Operation Overview*, all operations necessary to assemble your bike.

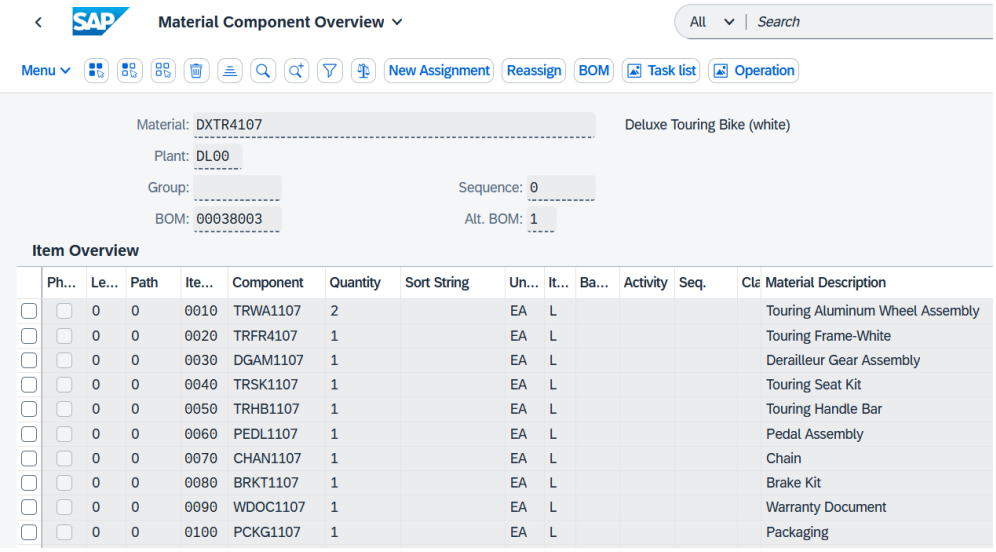
Enter



Ope...	SOp	Work Ce...	Plant	* C...	Standard...	Description	Lo...	PRT	Cl...	O...	Pe...	Cu...	Su...	Base Quantity
0010		ASSY1000	DL00	ASSY		Material staging								15
0020		ASSY1000	DL00	ASSY		Attach seat to frame								1
0030		ASSY1000	DL00	ASSY		Attach handle bar assembly								1
0040		ASSY1000	DL00	ASSY		Attach derailleur gear assm. to wheel								1
0050		ASSY1000	DL00	ASSY		Attach front and real wheels to chain								1
0060		ASSY1000	DL00	ASSY		Attach brakes								1
0070		ASSY1000	DL00	ASSY		Attach peddles								1
0080		INSP1000	DL00	ASSY		Test bike								1
0090		PACK1000	DL00	ASSY		Disassemble								1
0100		PACK1000	DL00	ASSY		Pack bike								1
0110		PACK1000	DL00	ASSY		Move to storage								15

Make sure that the modified BOM is already valid by selecting operation *0010* and clicking Allocation. If this option is not displayed, you will find it in the pull-down menu under *Menu* ► *Goto* ► *Allocation*.

Your previously created frame should be listed as *Item Number 20*.



Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.	Cl	Material Description
	0	0	0010	TRWA1107	2		EA	L					Touring Aluminum Wheel Assembly
	0	0	0020	TRFR4107	1		EA	L					Touring Frame-White
	0	0	0030	DGAM1107	1		EA	L					Derailleur Gear Assembly
	0	0	0040	TRSK1107	1		EA	L					Touring Seat Kit
	0	0	0050	TRHB1107	1		EA	L					Touring Handle Bar
	0	0	0060	PEDL1107	1		EA	L					Pedal Assembly
	0	0	0070	CHAN1107	1		EA	L					Chain
	0	0	0080	BRKT1107	1		EA	L					Brake Kit
	0	0	0090	WDOC1107	1		EA	L					Warranty Document
	0	0	0100	PCKG1107	1		EA	L					Packaging

Click on Save. You will get a confirmation from the sytem.



Click on SAP to return to the SAP Fiori Launchpad.

Step 6: Create Cost Estimate

Task Create a cost estimate for your bike. **Time** 10 min

Short Description Create a cost estimate for your white Deluxe Touring Bike and review the cost components.

Name (Position) Jamie Shamblin (Controller)

Product costing calculates the cost of goods manufactured and the cost of goods sold per product unit. Products are costed automatically using bills of materials and routings from Production Planning.

To estimate costs, use the *Create Material Cost Estimate* app in the *Controlling* area on the *Product Cost Controlling* page in the *Controller* role.

Start



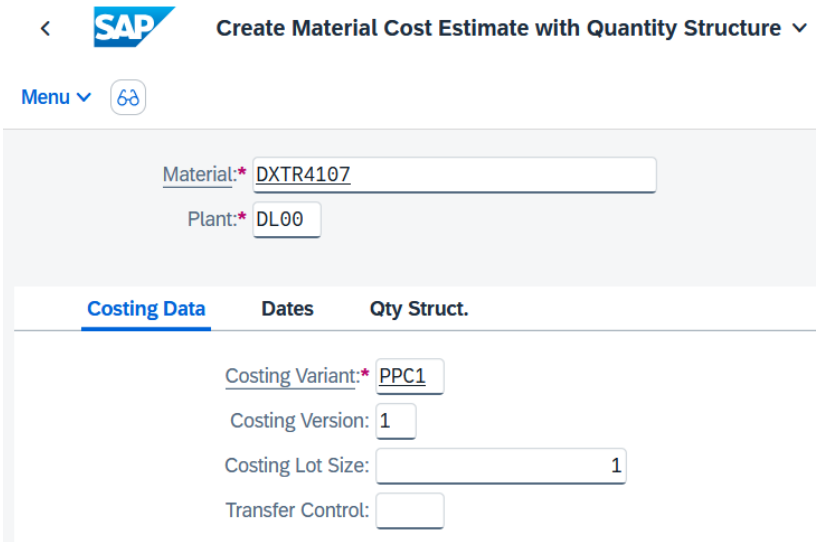
On the *Create Material Cost Estimate with Quantity Structure* screen, choose your bike **DXTR4###** and plant **DL00**.

DXTR4###
DL00

On the *Costing Data* tab, choose **Standard Cost Est. (PPC1)** as the *Costing Variant* via the , *Costing Version* **1** and **1** as the *Costing Lot Size*. Compare with the screen shown below.

Standard Cost Est.

1
1

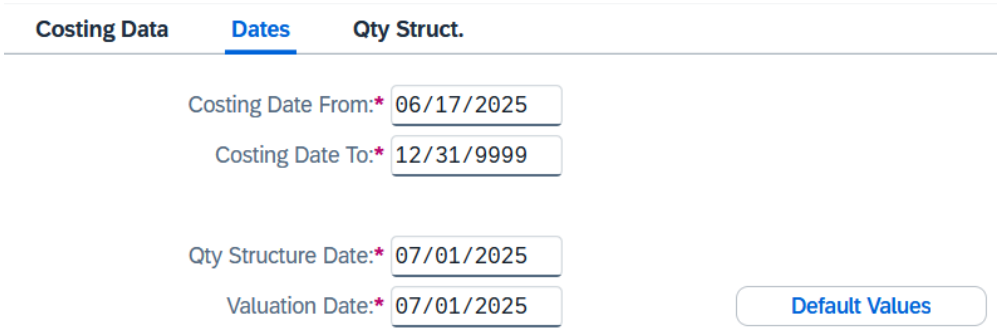


Press **Enter**. If a warning occurs close it and continue.

Enter

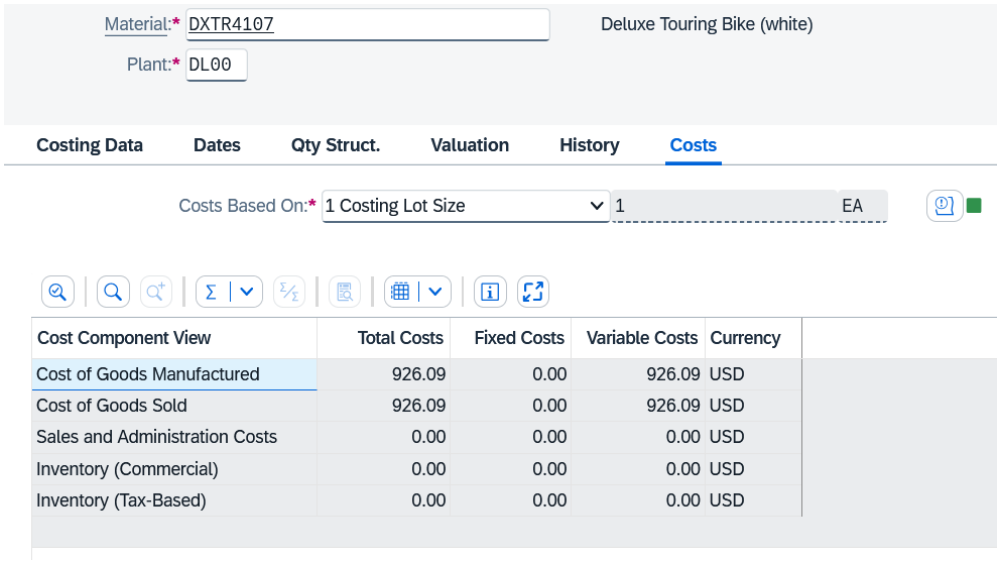
On the *Dates* tab, change the *Costing Date From* to the **current date** and press Enter to confirm all other date entries. The system now calculates the material based on the bills of material and the routings.

Current date



Press **Enter** to confirm all other entries. The system now calculates the costs based on the bill of materials and the routing.

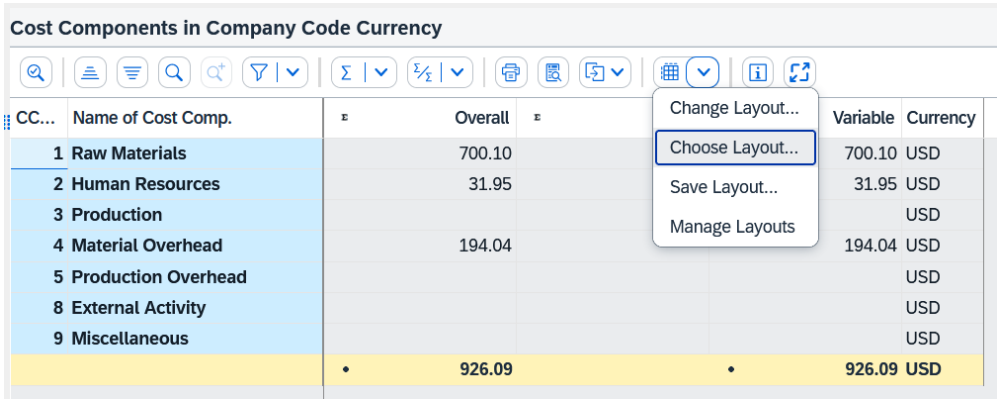
Enter



Material costs of 926.09 USD are calculated for the Deluxe Touring Bike.

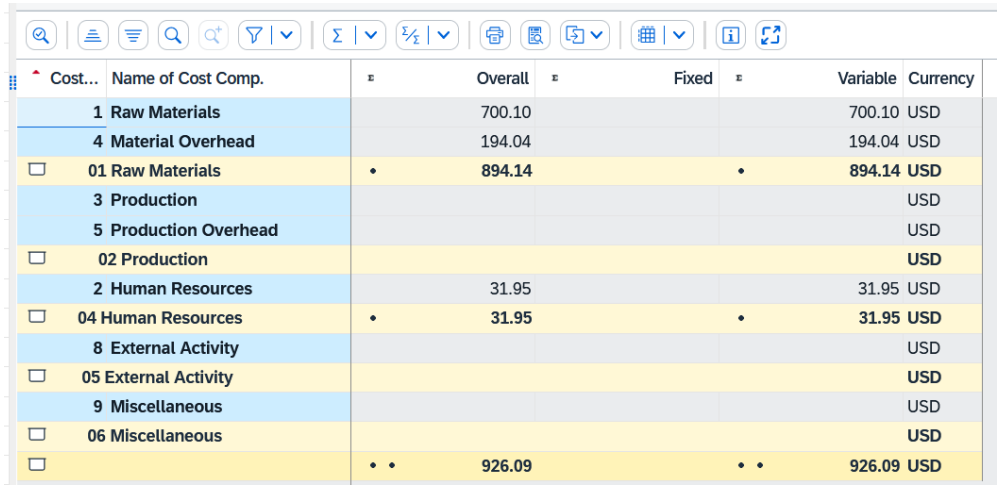
To display the exact cost components choose from the Pull-Down-Menu **Menu ▶ Costs ▶ Display Cost Components.**

Then select in the header area of the lower table and click *Choose Layout...*.



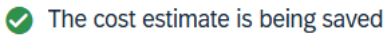
CC...	Name of Cost Comp.	Overall	Variable	Currency
1	Raw Materials	700.10	700.10	USD
2	Human Resources	31.95	31.95	USD
3	Production			USD
4	Material Overhead	194.04	194.04	USD
5	Production Overhead			USD
8	External Activity			USD
9	Miscellaneous			USD
		926.09	926.09	USD

Choose the Layout **1SAP02 - Cost Component Groups**. The table will be altered. 1SAP02



Cost...	Name of Cost Comp.	Overall	Fixed	Variable	Currency
1	Raw Materials	700.10		700.10	USD
4	Material Overhead	194.04		194.04	USD
01	Raw Materials	894.14		894.14	USD
3	Production				USD
5	Production Overhead				USD
02	Production				USD
2	Human Resources	31.95		31.95	USD
04	Human Resources	31.95		31.95	USD
8	External Activity				USD
05	External Activity				USD
9	Miscellaneous				USD
06	Miscellaneous				USD
		926.09		926.09	USD

Select Save to save your costing. You will get a confirmation from the system.



Click on SAP to return to the SAP Fiori Launchpad.

Step 7: Mark Price Updates

Task Mark the standard cost estimate. **Time** 5 min

Short Description Transfer the cost estimate to the material master record (future Price).

Name (Position) Jamie Shamblin (Controller)

The Price Update function is used to transfer the calculated product cost as a cost estimation to the material master record. The cost is transferred into two steps: Marking and release.

To mark prices, in the *Controlling* area, on the *Product Cost Controlling* page in the *Controller* role, use the *Release Material Cost Estimates* app.

Start



Enter the **current period** (current month), *company code* **US00**, *plant* **DL00** and your bike **DXTR4###** as *Material*.

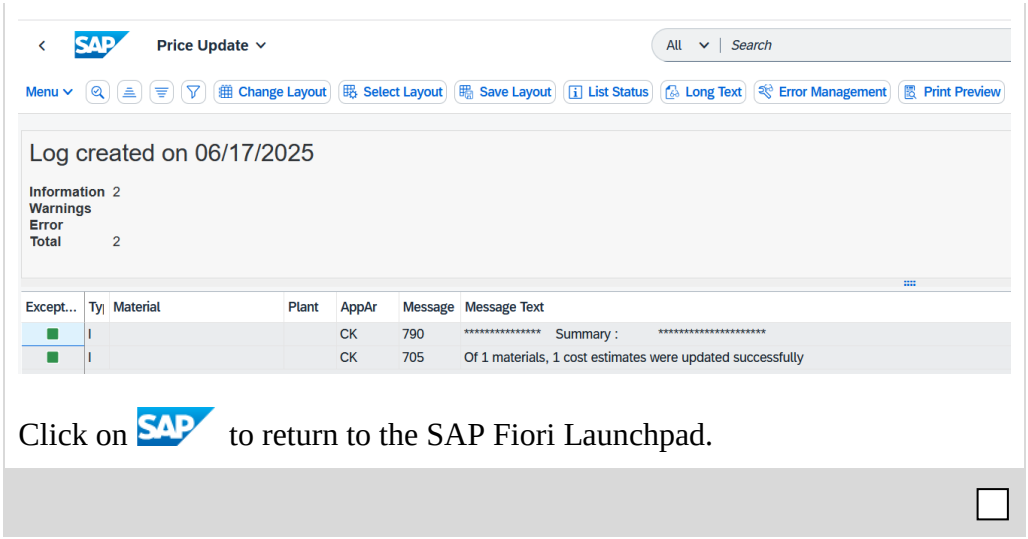
Current period
US00
DL00
DXTR4###



Remove the indicator from the **Test Run** field. Execute the marking by pressing Execute.

~~Test Run~~

The system should display the following message:



Click on SAP to return to the SAP Fiori Launchpad.

Step 8: View Prices Changes

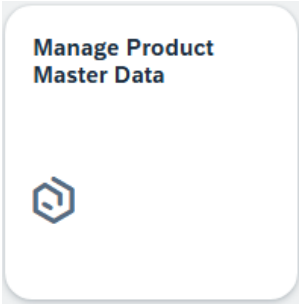
Task View the planned price. **Time** 5 min

Short Description View the planned price in the material master record of your white Deluxe Touring Bike.

Name (Position) Shuyuan Chen (Head of Accounting)

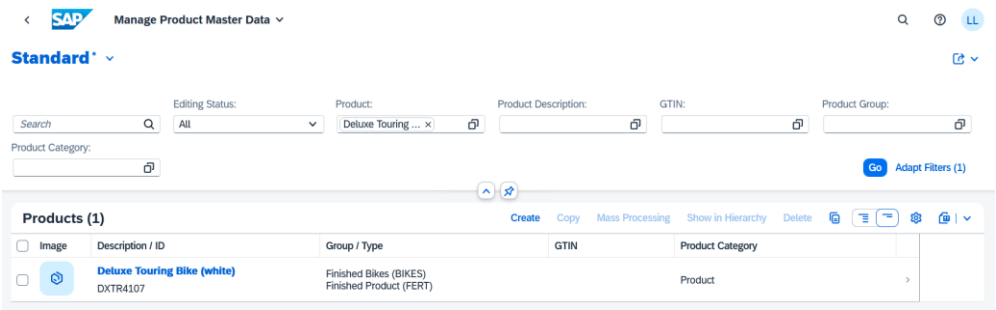
To view prices, in the *Controlling* area, on the *Product Cost Controlling* page, in the *Head of Accounting* role, use the *Manage Product Master Data* app.

Start



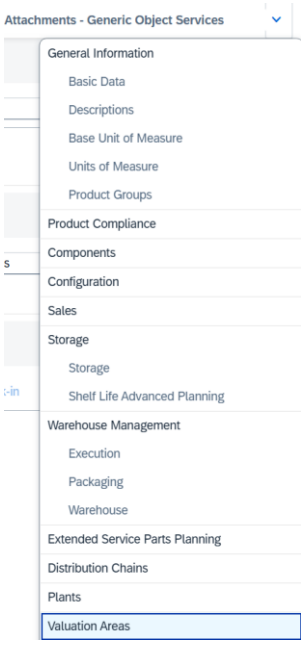
Enter **DXTR4###** in the *Product* field and klick on Go.

DXTR4###

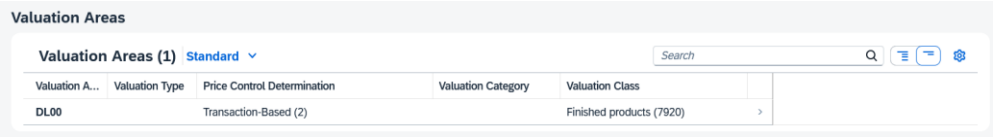


Click within the row of your only search result to open the product details. Use the pull-down menu to select the *Valuation Areas* section. The screen automatically scrolls to the appropriate section.

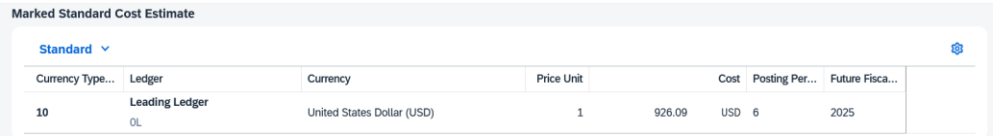
Valuation Areas



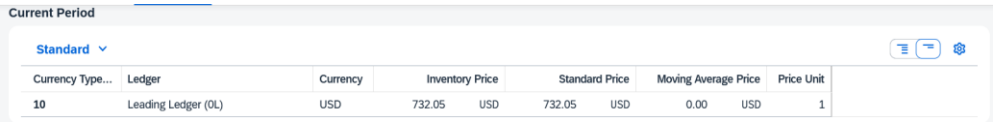
Choose the row with Valuation Area *DL00* to see further details.



Select the *Other Prices* tab, the screen automatically scrolls to it. On *Marked Standard Cost Estimate*, you will see a line with the details of the previously planned updated material cost estimate.



For comparison, select the tab *Valuation*, the screen automatically scrolls to it. On *Current Period*, you can see the currently valid standard price.



Click on SAP to return to the SAP Fiori Launchpad.

Step 9: Release Price Updates

Task Release the price updates. **Time** 5 min

Short Description Release the price updates.

Name (Position) Shuyuan Chen (Head of Accounting)

To update the price, in the *Controlling* area, on the *Product Cost Controlling* page, in the *Head of Accounting* role, use the *Release Material Cost Estimates* app.

Start



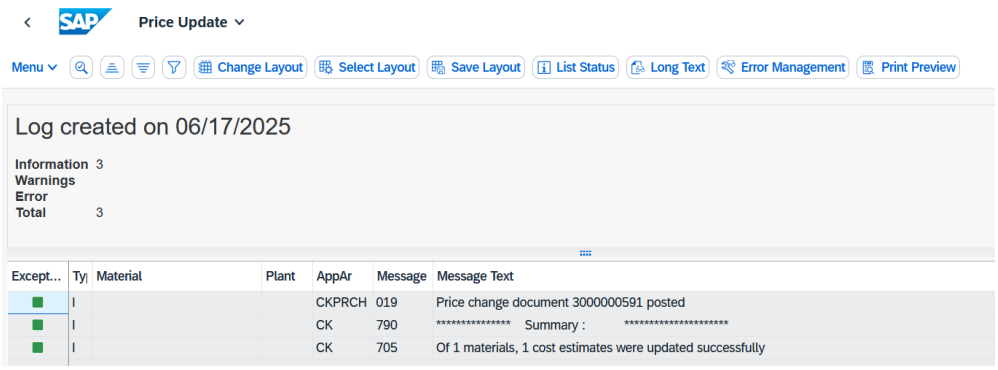
Select Release. Choose the **current period**, *Company Code* **US00**, *Plant* **DL00** and *Material* **DXTR4###**.

Current period, US00
DL00, DXTR4###

Remove the Test Run indicator and release the price by selecting Execute.

~~Test Run~~

The price change has now been posted. The system should display the following message:



Click on SAP to return to the SAP Fiori Launchpad.

Step 10: View Prices

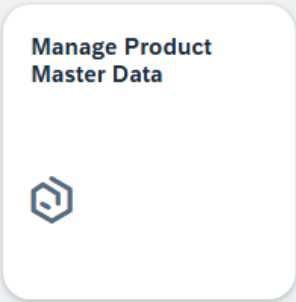
Task Review the current prices. **Time** 5 min

Short Description View the current prices.

Name (Position) Shuyuan Chen (Head of Accounting)

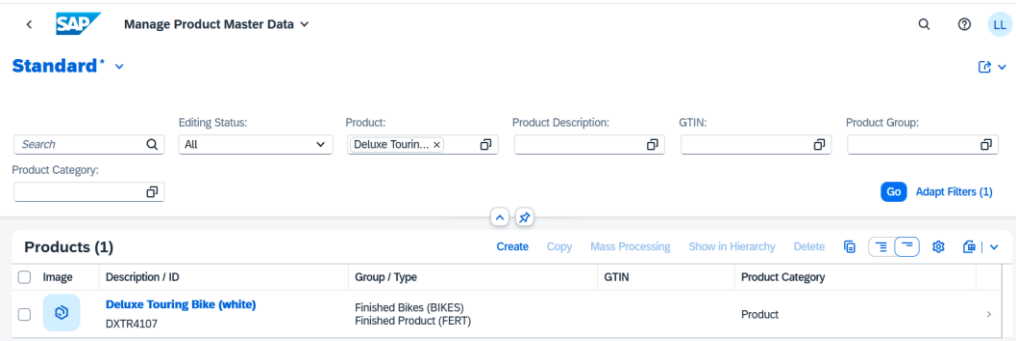
To view prices, in the *Controlling* area, on the *Product Cost Controlling* page, in the *Head of Accounting* role, use the *Manage Product Master Data* app.

Start



Enter **DXTR4###** in the *Product* field and klick on Go.

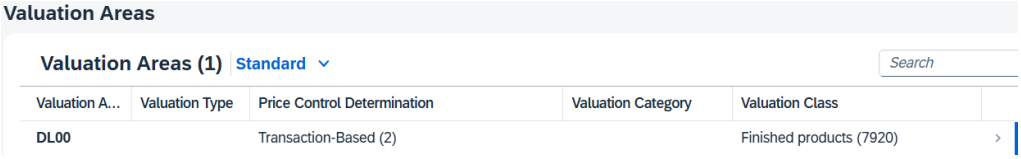
DXTR4###



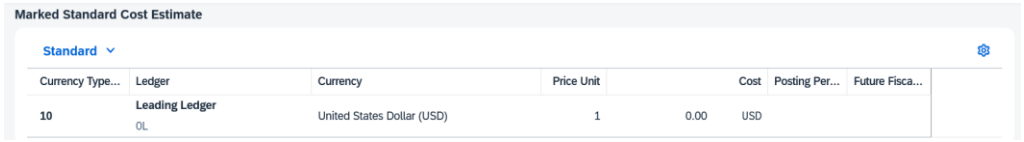
Click within the row of your only search result to open the product details. Use the pull-down menu to select the *Valuation Areas* section. The screen automatically scrolls to the appropriate section.

Valuation Areas

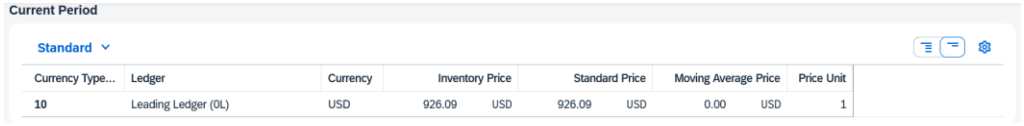
Choose the row with Valuation Area *DL00* to see further details.



Select the *Other Prices* tab, the screen automatically scrolls to it. On *Marked Standard Cost Estimate*, you will see that there is no more cost estimation present, since it is already posted.



For comparison, select the tab *Valuation*, the screen automatically scrolls to it. On *Current Period*, you can see the currently valid standard price.



Click on  to return to the SAP Fiori Launchpad.

CO-PC I Challenge

Learning Objective Understand and perform an integrated product costing process.

Time 45 min

Motivation After you have successfully worked through the *Product Costing* case study, you should be able to solve the following task on your own.

Scenario In the case study you have personalized the routing and the bill of material for the white Deluxe Touring Bike. Some customers have complained because of bad quality of the tires. For this reason, from next month on two touring tires and two touring tubes respectively should be additionally delivered. This will obviously change the product costing. Calculate the new price onwards so that it will be valid from next month on.

Task Information Since this task is based on the *Product Costing (CO-PC I)* case study, you can use it as guidance. However, it is recommended that you solve it without any help in order to test your acquired knowledge.

